

Application Deployment

(Deploy the given react application to a production ready state)

Application:

Clone the below mentioned repo and deploy the application.
(Run the application in port 80 [HTTP])

Repo URL: <https://github.com/sriram-R-krishnan/devops-build>

Docker:

- Dockerize the application by creating a Dockerfile
- Create a docker-compose file to use the above image

Bash Scripting:

- Write 2 scripts
 - **build.sh** for building docker images
 - **deploy.sh** for deploying the image to server

Version Control:

- Push the code to github to **dev branch** (use **dockerignore** & **gitignore** files)
- **Note:** Use only CLI for related git commands

Docker hub:

- Create 2 repos "**dev**" and "**prod**" to push images.
- "**Prod**" repo must be private and "**dev**" repo can be public

Jenkins:

- Install and configure jenkins build step as per needs to build, push & deploy the application
- Connect jenkins to the github repo with auto build trigger from both **dev** & **master** branch
- If code pushed to **dev branch**, docker image must build and pushed to **dev repo** in docker hub
- If **dev merged to master**, then docker image must be pushed to **prod repo** in docker hub

AWS:

- Launch **t2.micro instance** and deploy the create application.
- Configure SG as below:
 - Whoever has the ip address can access the application
 - Login to server can should be made only from your ip address

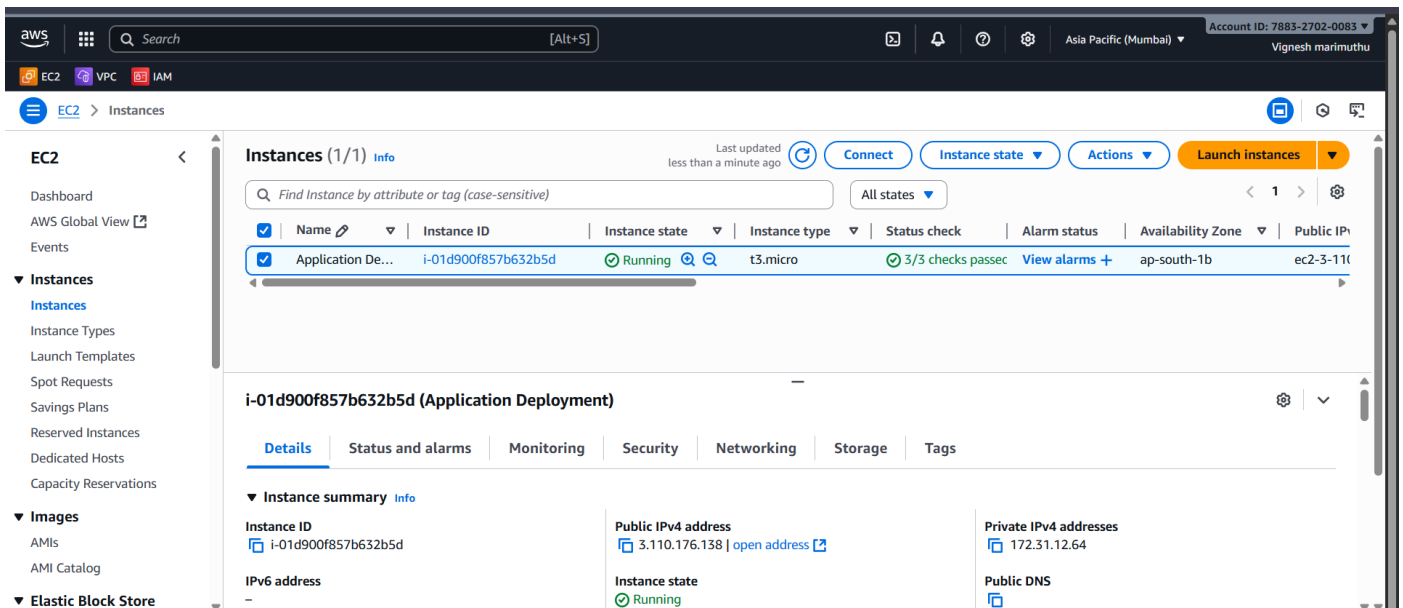
Monitoring:

- Setup a monitoring system to check the health status of the application. (Open-source)
- Sending notifications only if the application goes down is highly appreciable

Solution:

✓ 1. Launched an EC2 instance and connected to Ubuntu via SSH

- EC2 will act as the *Jenkins build server and control center* for CI/CD, Docker builds, etc.



```
vignesh@Vignesh:~/ssh$ ssh -i ~/.ssh/3_10_key.pem ubuntu@3.110.176.138
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Oct 11 18:56:14 UTC 2025

System load: 0.03          Temperature: -273.1 C
Usage of /: 9.4% of 18.33GB Processes: 118
Memory usage: 24%         Users logged in: 0
Swap usage: 0%            IPv4 address for ens5: 172.31.12.64
```

✓ 2. Prepare the React repository (locally)

```
ubuntu@ip-172-31-12-64:~$ git clone https://github.com/sriram-R-krishnan/devops-build
Cloning into 'devops-build'...
remote: Enumerating objects: 21, done.
remote: Total 21 (delta 0), reused 0 (delta 0), pack-reused 21 (from 1)
Receiving objects: 100% (21/21), 720.09 KiB | 17.56 MiB/s, done.
ubuntu@ip-172-31-12-64:~/devops-build$
```

Created .dockerignore & .gitignore

- .gitignore avoids huge or secret files being committed.
- .dockerignore avoids sending unnecessary files to the Docker daemon during build.

```
ubuntu@ip-172-31-12-64:~/devops-build$ nano .dockerignore
ubuntu@ip-172-31-12-64:~/devops-build$ cat .dockerignore
.git
.gitignore
node_modules
Dockerfile
docker-compose.yml
ubuntu@ip-172-31-12-64:~/devops-build$ nano .gitignore
ubuntu@ip-172-31-12-64:~/devops-build$ cat .gitignore
node_modules
build
.env
ubuntu@ip-172-31-12-64:~/devops-build$
```

Created Dockerfile

- Used a lightweight web server image (Nginx) and copy only the built static files into it.
- This makes our final image very small and secure.

```
ubuntu@ip-172-31-12-64:~/devops-build$ nano Dockerfile
ubuntu@ip-172-31-12-64:~/devops-build$ cat Dockerfile
# Use Nginx to serve the pre-built React app
FROM nginx:1.23-alpine

# Copy existing React build to Nginx web directory
COPY build /usr/share/nginx/html

# Expose port 80
EXPOSE 80

# Start Nginx
CMD ["nginx", "-g", "daemon off;"]
ubuntu@ip-172-31-12-64:~/devops-build$
```

Created docker-compose.yml

- docker-compose up -d provides convenient local testing

```
ubuntu@ip-172-31-12-64:~/devops-build$ nano docker-compose.yml
ubuntu@ip-172-31-12-64:~/devops-build$ cat docker-compose.yml
version: '3.8'
services:
  web:
    build: .
    container_name: react_app
    ports:
      - "80:80"
    restart: always
ubuntu@ip-172-31-12-64:~/devops-build$
```

```
ubuntu@ip-172-31-12-64:~/devops-build$ docker-compose up -d
Creating network "devops-build_default" with the default driver
Building web
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 2.621MB
Step 1/9 : FROM node:18-alpine AS builder
18-alpine: Pulling from library/node
f18232174bc9: Pulling fs layer
dd71dde834b5: Pulling fs layer
1e5a4c89cee5: Pulling fs layer
25ff2da83641: Pulling fs layer
25ff2da83641: Waiting
f18232174bc9: Verifying Checksum
f18232174bc9: Download complete
f18232174bc9: Pull complete
1e5a4c89cee5: Verifying Checksum
1e5a4c89cee5: Download complete
25ff2da83641: Verifying Checksum
25ff2da83641: Download complete
```

```
Step 4/4 : CMD ["nginx", "-g", "daemon off;"]
--> Running in 98b9331e2971
--> Removed intermediate container 98b9331e2971
--> 094c0b662020
Successfully built 094c0b662020
Successfully tagged devops-build_web:latest
Creating react_app ... done
ubuntu@ip-172-31-12-64:~/devops-build$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
62ba80fe0ff2   devops-build_web  "/docker-entrypoint.…"   6 seconds ago  Up 6 seconds  0.0.0.0:80->80/tcp, [::]:80->80/tcp  react_app
ubuntu@ip-172-31-12-64:~/devops-build$
```

✓ 3. Added build & deploy scripts

- These scripts will be used later by Jenkins.
- Make the scripts executable.

```
ubuntu@ip-172-31-12-64:~/devops-build$ cat build.sh
#!/bin/bash
if [ -z "$1" ]; then
    TAG="latest"
else
    TAG="$1"
fi
docker build -t $TAG .
ubuntu@ip-172-31-12-64:~/devops-build$ nano deploy.sh
ubuntu@ip-172-31-12-64:~/devops-build$ cat deploy.sh
#!/bin/bash
IMAGE="$1"
CONTAINER="react_app"

# stop old container (if running)
docker ps -q -f name=$CONTAINER | grep -q . && docker stop $CONTAINER && docker rm $CONTAINER || echo "no container to stop"

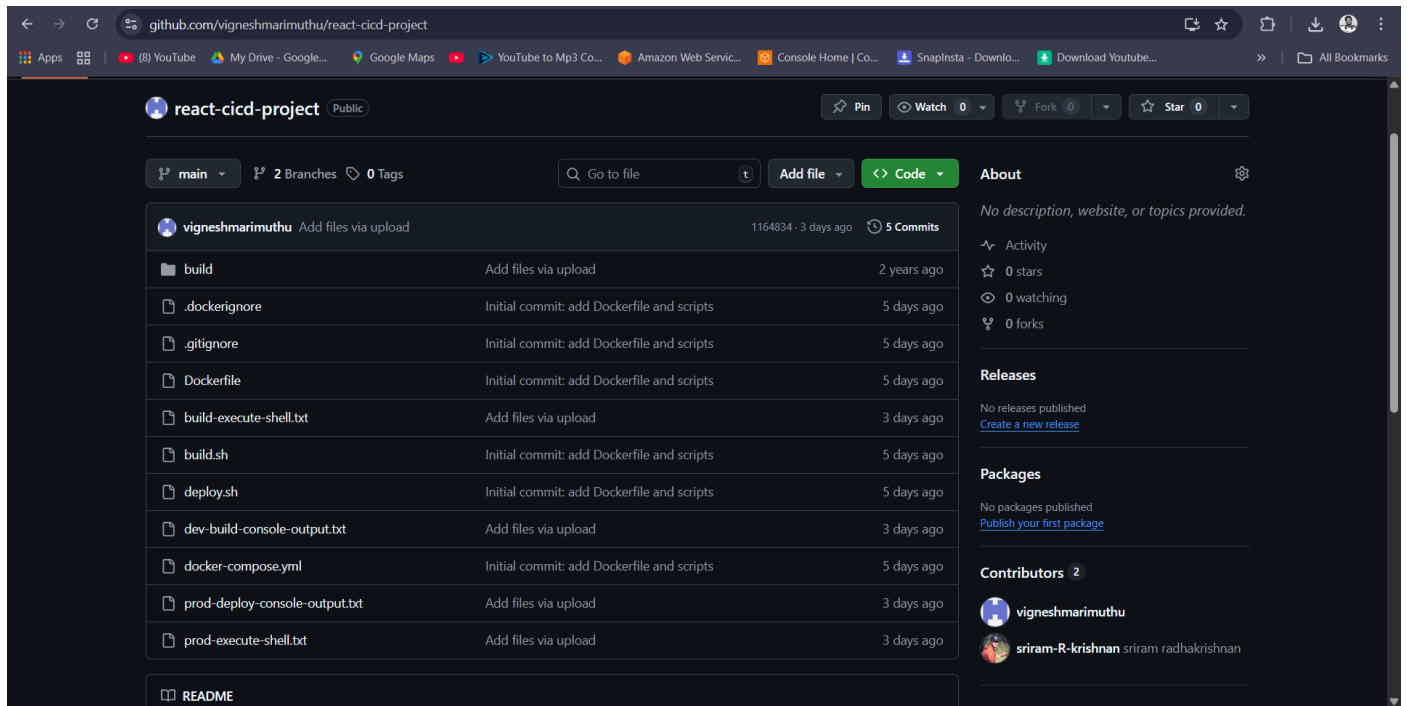
# pull & run
docker pull $IMAGE
docker run -d -p 80:80 --name $CONTAINER $IMAGE
ubuntu@ip-172-31-12-64:~/devops-build$ chmod +x build.sh deploy.sh
ubuntu@ip-172-31-12-64:~/devops-build$
```

✓ 4. Git branches: created dev and main (locally) and push to my GitHub repo

- **dev** branch holds work-in-progress; **main** (or master) is production branch. Having both enables the pipeline behavior requested.

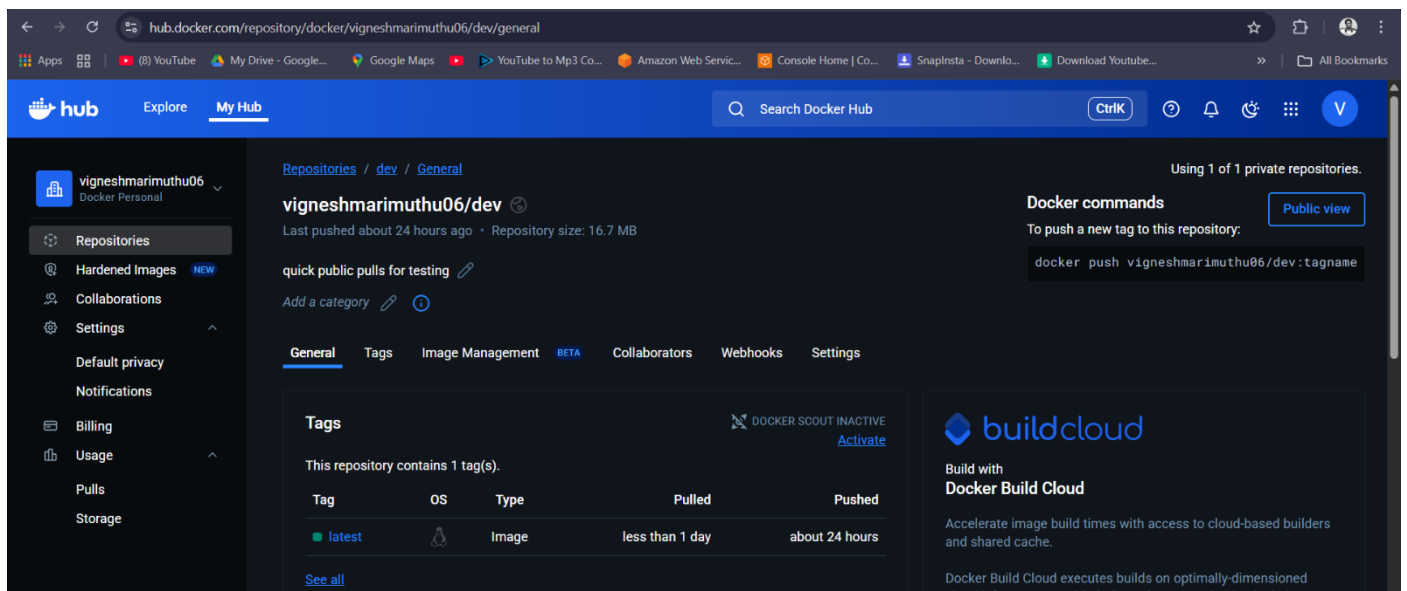
```
ubuntu@ip-172-31-12-64:~/devops-build$ git remote -v
origin https://github.com/sriram-R-krishnan/devops-build (fetch)
origin https://github.com/sriram-R-krishnan/devops-build (push)
ubuntu@ip-172-31-12-64:~/devops-build$ git remote remove origin
ubuntu@ip-172-31-12-64:~/devops-build$ git remote add origin https://github.com/vigneshmarimuthu/react-cicd-project.git
ubuntu@ip-172-31-12-64:~/devops-build$ git push -u origin main
git push -u origin dev
Username for 'https://github.com': vigneshmarimuthu
Password for 'https://vigneshmarimuthu@github.com':
Enumerating objects: 30, done.
Counting objects: 100% (30/30), done.
Delta compression using up to 2 threads
Compressing objects: 100% (27/27), done.
Writing objects: 100% (30/30), 721.35 KiB | 120.22 MiB/s, done.
Total 30 (delta 0), reused 21 (delta 0), pack-reused 0
To https://github.com/vigneshmarimuthu/react-cicd-project.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.
Username for 'https://github.com': vigneshmarimuthu
Password for 'https://vigneshmarimuthu@github.com':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'dev' on GitHub by visiting:
remote:   https://github.com/vigneshmarimuthu/react-cicd-project/pull/new/dev
remote:
To https://github.com/vigneshmarimuthu/react-cicd-project.git
 * [new branch]    dev -> dev
branch 'dev' set up to track 'origin/dev'.
ubuntu@ip-172-31-12-64:~/devops-build$
```

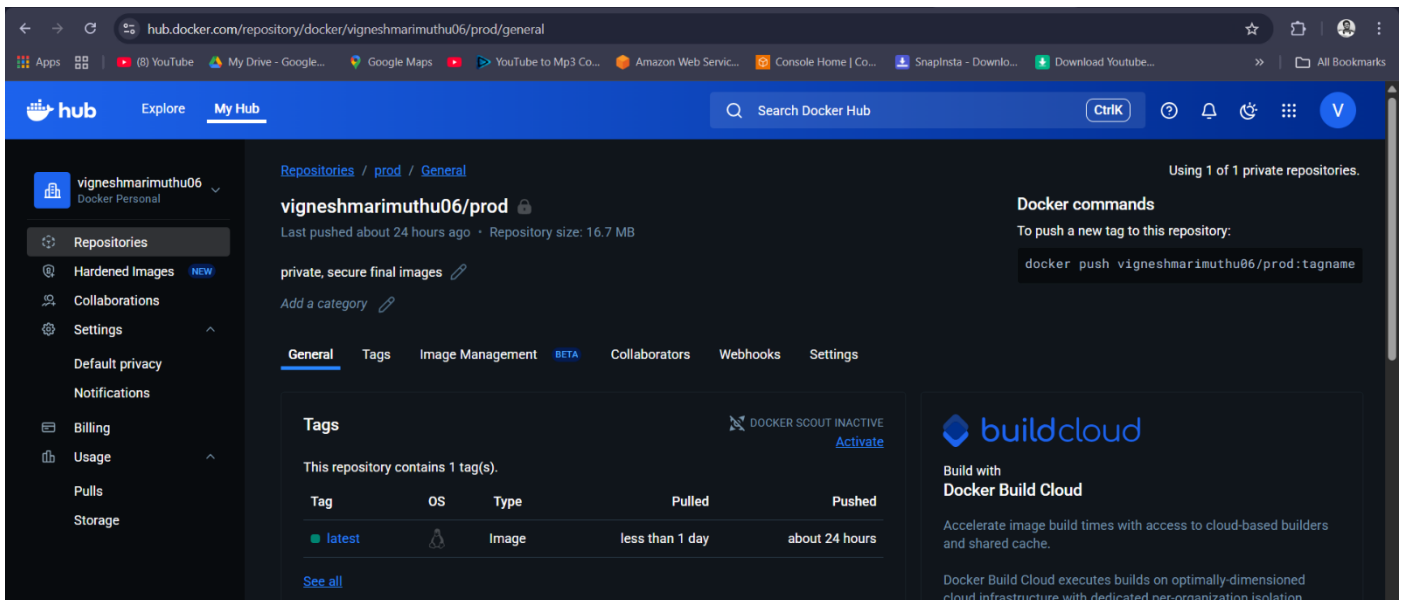
- My GitHub repository: <https://github.com/vigneshmarimuthu/react-cicd-project>



✓ 5. Create Docker Hub repositories

- Dev repo: quick public pulls for testing. Prod repo: private, secure final images.





✓ 6. Set up the EC2 Ubuntu server: install Docker and Docker Compose

- Jenkins (running in a container) will use the host Docker socket to build images.

```
ubuntu@ip-172-31-12-64:~/devops-build$ sudo apt-get update -y
sudo apt-get upgrade -y
sudo apt-get install -y ca-certificates curl gnupg lsb-release
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
systemd-hwe-hwdb
```

```
ubuntu@ip-172-31-12-64:~/devops-build$ # add Docker's official GPG key
sudo mkdir -p /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg

# set up repository
echo \
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] https://download.docker.com/linux/ubuntu \
$(lsb_release -cs) stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

sudo apt-get update -y
sudo apt-get install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:6 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [33.0 kB]
```

```
ubuntu@ip-172-31-12-64:~/devops-build$ sudo usermod -aG docker $USER
ubuntu@ip-172-31-12-64:~/devops-build$ exit
logout
Connection to 3.110.176.138 closed.
vignesh@Vignesh:~/.ssh$ ssh -i ~/.ssh/3_10_key.pem ubuntu@3.110.176.138
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Oct 11 19:21:51 UTC 2025

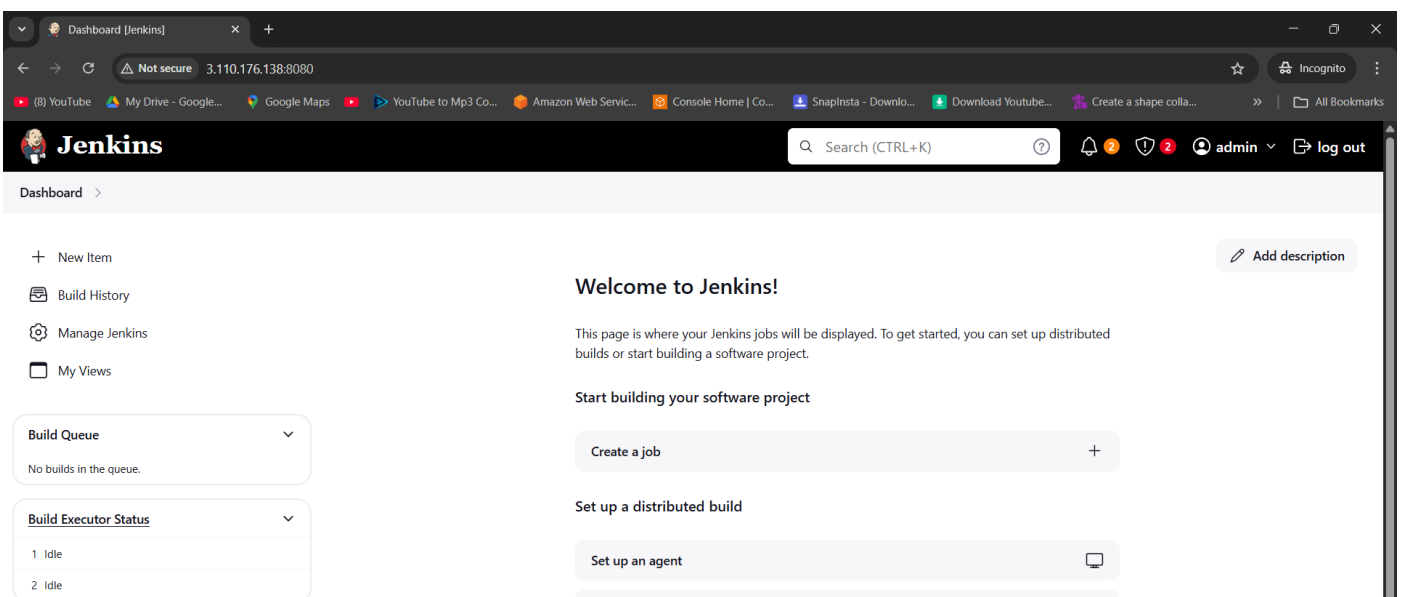
System load: 0.39           Temperature: -273.1 C
Usage of /: 19.2% of 18.3GB Processes: 116
Memory usage: 33%          Users logged in: 0
Swap usage: 0%             IPv4 address for ens5: 172.31.12.64
```

✅ 7. Installed and run Jenkins (in Docker) on EC2

- `-v /var/run/docker.sock:/var/run/docker.sock` lets Jenkins control the host Docker (build/push images) from inside container.
- This is standard and widely used, but it grants elevated access to Docker on the host — secure the server.

```
ubuntu@ip-172-31-12-64:~$ docker run -d --name jenkins \
-p 8080:8080 -p 50000:50000 \
-v ~/jenkins_home:/var/jenkins_home \
-v /var/run/docker.sock:/var/run/docker.sock \
jenkins/jenkins:lts-jdk11
Unable to find image 'jenkins/jenkins:lts-jdk11' locally
lts-jdk11: Pulling from jenkins/jenkins
cdd62bf39133: Pull complete
21f106ffc421: Pull complete
39df2c5808cf: Pull complete
d9d5ad5daae2: Pull complete
21d9152ebad0: Pull complete
ddc06df74615: Pull complete
bf388b3d4868: Pull complete
1f6fc1ff002b: Pull complete
6201e887d163: Pull complete
a3d60a50862f: Pull complete
282c783b8e01: Pull complete
ddf03cc24103: Pull complete
Digest: sha256:6aa6c6bd7da914bf5333305c8102cb26965ea4b227e37f4269315725a2b0cd81
Status: Downloaded newer image for jenkins/jenkins:lts-jdk11
0dba0b44cb9d9bb188c912d9b3f99cdd27087f6f8fe38d35d7f15ca99daba1eb
ubuntu@ip-172-31-12-64:~$ docker ps
CONTAINER ID   IMAGE                                COMMAND                                  CREATED      STATUS      PORTS
0dba0b44cb9d   jenkins/jenkins:lts-jdk11           "/usr/bin/tini -- /u..."             45 seconds ago Up 44 seconds 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
62ba80fe0ff2   devops-build_web                    "/docker-entrypoint..."             17 minutes ago Up 2 minutes  0.0.0.0:80->80/tcp, [::]:80->80/tcp
ubuntu@ip-172-31-12-64:~$
```

✅ 8. Opened Jenkins UI



The screenshot shows the Jenkins web interface in a browser. The address bar indicates the URL is `3.110.176.138:8080`. The page header includes the Jenkins logo, a search bar, and the user `admin` with a `log out` button. The main content area features a sidebar on the left with links to `New Item`, `Build History`, `Manage Jenkins`, and `My Views`. The central panel displays a `Welcome to Jenkins!` message, explaining that this is where Jenkins jobs are managed. It includes a section titled `Start building your software project` with a `Create a job` button, and another section titled `Set up a distributed build` with a `Set up an agent` button. On the left sidebar, there are also sections for `Build Queue` (showing no builds) and `Build Executor Status` (showing two idle executors).

Jenkins: added credentials (Docker Hub)

The screenshot shows the Jenkins web interface for creating new credentials. The breadcrumb trail is: Dashboard > Manage Jenkins > Credentials > System > Global credentials (unrestricted). The form is titled 'New credentials' and includes the following fields:

- Kind:** A dropdown menu set to 'Username with password'.
- Scope:** A dropdown menu set to 'Global (Jenkins, nodes, items, all child items, etc)'.
- Username:** A text input field containing 'vigneshmarimuthu06'.
- Treat username as secret:** An unchecked checkbox.
- Password:** A password input field with masked characters (dots).
- ID:** A text input field containing 'dockerhub'.

A blue 'Create' button is located at the bottom left of the form.

Configure GitHub webhook → Jenkins

The screenshot shows the GitHub repository settings page for 'vigneshmarimuthu / react-cicd-project'. The 'Settings' tab is selected, and the 'Webhooks / Manage webhook' section is active. The 'Settings' sub-tab is chosen, showing the following configuration:

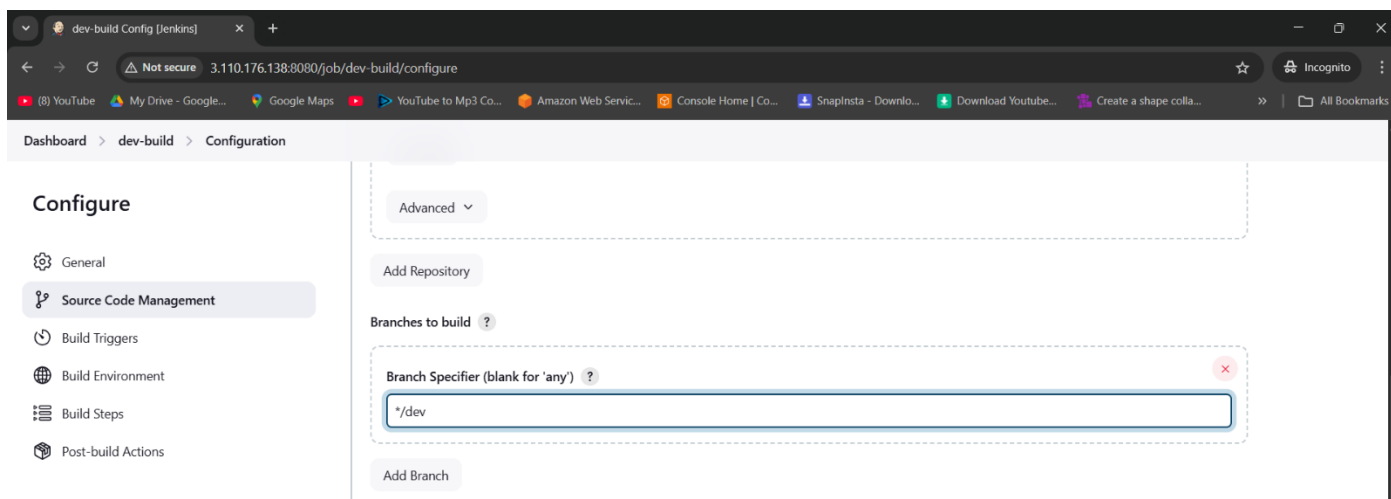
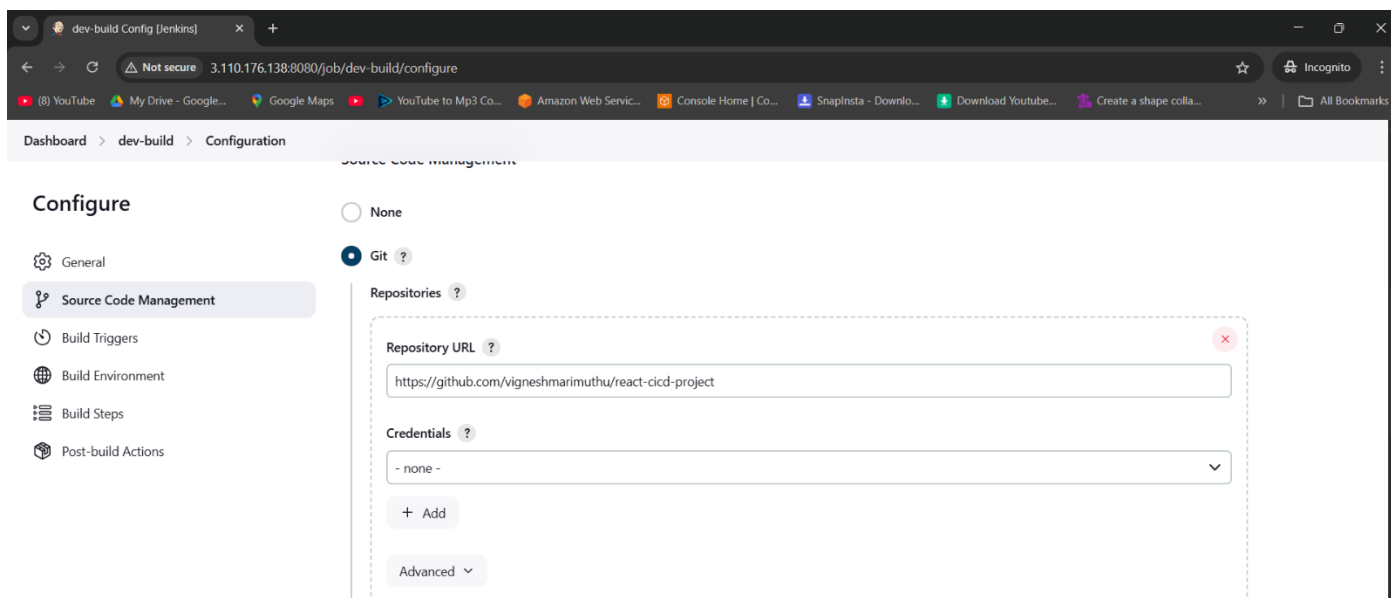
- Settings / Recent Deliveries:** Two tabs are visible, with 'Settings' currently selected.
- Instructions:** A text block explaining that a POST request will be sent to the specified URL with details of subscribed events. It mentions that the format can be JSON, x-www-form-urlencoded, etc., and refers to the developer documentation.
- Payload URL *:** A text input field containing 'http://3.110.176.138:8080/github-webhook/'.
- Content type *:** A dropdown menu set to 'application/json'.

A 'Preview' button is located at the bottom of the settings panel.

✓ 9. Created Jenkins Jobs (Freestyle)

1.dev-build (triggered on pushes to dev branch)

- Build the image from dev branch and push to vigneshmarimuthu06/dev (public). Used for testing.



dev-build Config [Jenkins]

Not secure 3.110.176.138:8080/job/dev-build/configure

Incognito

Dashboard > dev-build > Configuration

Configure

General

Source Code Management

Build Triggers

Build Environment

Build Steps

Post-build Actions

repository browser

(Auto)

Additional Behaviours

Add

Build Triggers

☐ Trigger builds remotely (e.g., from scripts)

☐ Build after other projects are built

☐ Build periodically

☒ GitHub hook trigger for GITScm polling

☐ Poll SCM

dev-build Config [Jenkins]

Not secure 3.110.176.138:8080/job/dev-build/configure

Incognito

Dashboard > dev-build > Configuration

Configure

General

Source Code Management

Build Triggers

Build Environment

Build Steps

Post-build Actions

Build Environment

☐ Delete workspace before build starts

☒ Use secret text(s) or file(s)

Bindings

Username and password (separated)

Username Variable

DOCKER_USER

Password Variable

DOCKER_PAS\$

Credentials

Specific credentials

Parameter expression

vigneshmarimuthu06/*****

dev-build Config [Jenkins]

Not secure 3.110.176.138:8080/job/dev-build/configure

Incognito

Dashboard > dev-build > Configuration

Configure

General

Source Code Management

Build Triggers

Build Environment

Build Steps

Post-build Actions

Execute shell ?

Command

See [the list of available environment variables](#)

```
#!/bin/bash
set -e

echo "Login to Docker Hub"
echo "$DOCKER_PASS" | docker login -u "$DOCKER_USER" --password-stdin

echo "Build Docker image"
docker build -t "$DOCKER_USER/dev:latest" .

echo "Push to Docker Hub (dev)"
docker push "$DOCKER_USER/dev:latest"

echo "✅ Build and push completed successfully!"
```

Advanced ▾

Save

Apply

dev-build #3 [Jenkins]

Not secure 3.110.176.138:8080/job/dev-build/3/

Incognito

Jenkins

Search (CTRL+K)

admin

log out

Dashboard > dev-build > #3

Status

Changes

Console Output

Edit Build Information

Delete build '#3'

Timings

Git Build Data

Previous Build

✓ #3 (Oct 11, 2025, 7:49:54 PM)

Add description

Keep this build forever

Started by user [admin](#)

Started 28 sec ago
Took **19 sec**

This run spent:

- 7 ms waiting;
- 19 sec build duration;
- 19 sec total from scheduled to completion.

git

Revision: 30ce8ddf0b3e0c31f11a85f589fef157e5eddd59

Repository: <https://github.com/vigneshmarimuthu/react-cicd-project>

- refs/remotes/origin/dev

</>

No changes.

dev-build #3 Console [Jenkins] x +

Not secure 3.110.176.138:8080/job/dev-build/3/console

Jenkins

Search (CTRL+K)

admin log out

Dashboard > dev-build > #3 > Console Output

Status

Changes

Console Output

Edit Build Information

Delete build '3'

Timings

Git Build Data

Previous Build

Console Output

Started by user [admin](#)

Running as SYSTEM

Building in workspace /var/jenkins_home/workspace/dev-build

The recommended git tool is: NONE

No credentials specified

> git rev-parse --resolve-git-dir /var/jenkins_home/workspace/dev-build/.git # timeout=10

Fetching changes from the remote Git repository

> git config remote.origin.url <https://github.com/vigneshmarimuthu/react-cicd-project> # timeout=10

Fetching upstream changes from <https://github.com/vigneshmarimuthu/react-cicd-project>

> git --version # timeout=10

> git --version # 'git version 2.39.5'

> git fetch --tags --force --progress -- <https://github.com/vigneshmarimuthu/react-cicd-project> +refs/heads/*:refs/remotes/origin/* # timeout=10

> git rev-parse refs/remotes/origin/dev^{commit} # timeout=10

Checking out Revision 30ce8ddf0b3e0c31f11a85f589fef157e5eddd59 (refs/remotes/origin/dev)

> git config core.sparsecheckout # timeout=10

> git checkout -f 30ce8ddf0b3e0c31f11a85f589fef157e5eddd59 # timeout=10

Commit message: "test: trigger dev pipeline"

> git rev-list --no-walk 30ce8ddf0b3e0c31f11a85f589fef157e5eddd59 # timeout=10

[dev-build] \$ /bin/bash /tmp/jenkins6783285226818710975.sh

Login to Docker Hub

- dev-build-execute-shell

```
#!/bin/bash
set -e

echo "Login to Docker Hub"
echo "$DOCKER_PASS" | docker login -u "$DOCKER_USER" --password-stdin

echo "Build Docker image"
docker build -t "$DOCKER_USER/dev:latest" .

echo "Push to Docker Hub (dev)"
docker push "$DOCKER_USER/dev:latest"

echo "✅ Build and push completed successfully!"
```

- I have downloaded dev-build-console-output

```
Started by user admin
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/dev-build
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/jenkins_home/workspace/dev-build/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/vigneshmarimuthu/react-cicd-project # timeout=10
Fetching upstream changes from https://github.com/vigneshmarimuthu/react-cicd-project
> git --version # timeout=10
> git --version # 'git version 2.39.5'
> git fetch --tags --force --progress -- https://github.com/vigneshmarimuthu/react-cicd-project +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/dev^{commit} # timeout=10
Checking out Revision 30ce8ddf0b3e0c31f11a85f589fef157e5eddd59 (refs/remotes/origin/dev)
> git config core.sparsecheckout # timeout=10
> git checkout -f 30ce8ddf0b3e0c31f11a85f589fef157e5eddd59 # timeout=10
Commit message: "test: trigger dev pipeline"
> git rev-list --no-walk 30ce8ddf0b3e0c31f11a85f589fef157e5eddd59 # timeout=10
[dev-build] $ /bin/bash /tmp/jenkins6783285226818710975.sh
Login to Docker Hub
```

WARNING! Your credentials are stored unencrypted in '/var/jenkins_home/.docker/config.json'.
Configure a credential helper to remove this warning. See
<https://docs.docker.com/go/credential-store/>

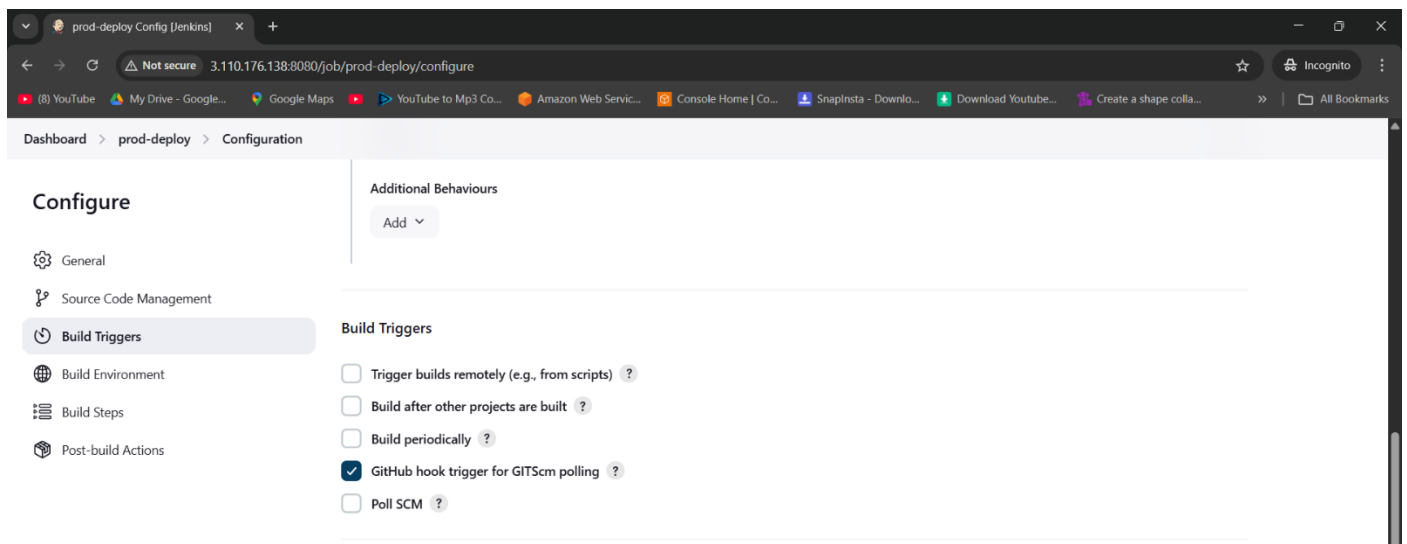
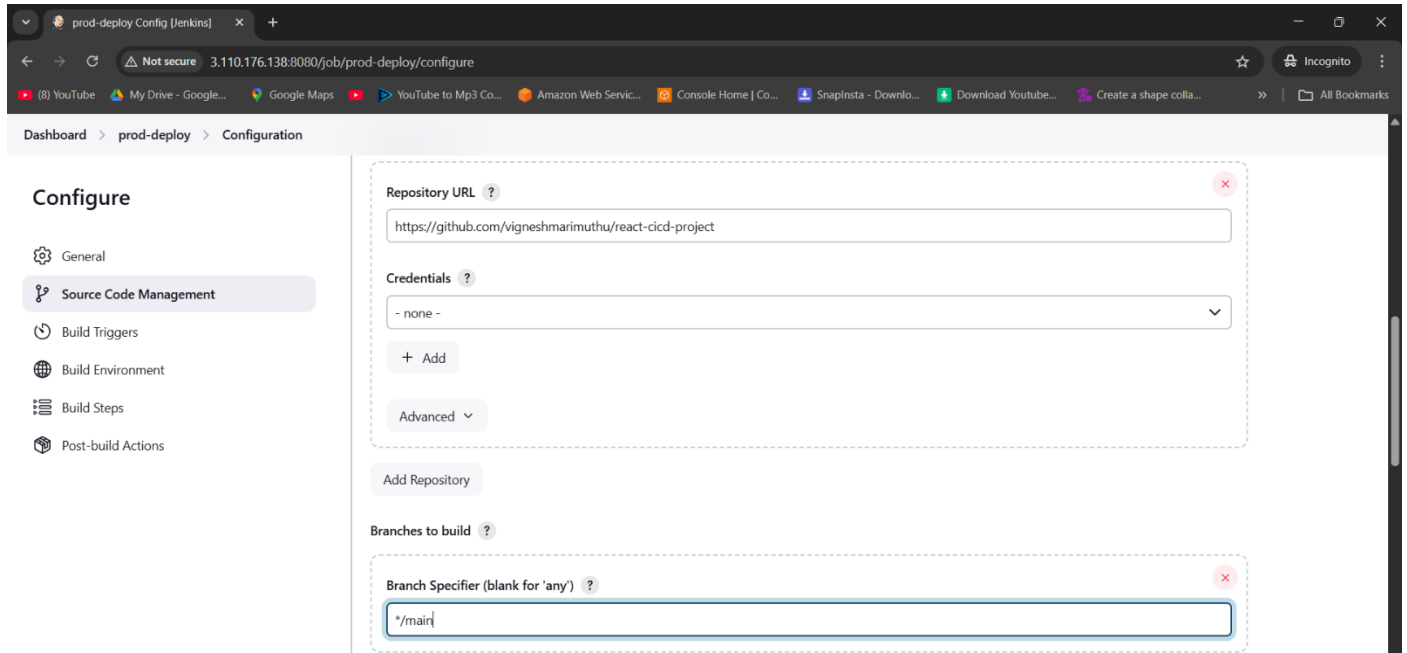
```
Login Succeeded
Build Docker image
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/
```

```
Sending build context to Docker daemon 2.625MB
Step 1/4 : FROM nginx:1.23-alpine
--> 8e75cbc5b25c
Step 2/4 : COPY build /usr/share/nginx/html
--> 754815867973
Step 3/4 : EXPOSE 80
--> Running in 5261a5d817d3
--> Removed intermediate container 5261a5d817d3
--> 80065fec5be8
```

```
Sending build context to Docker daemon 2.625MB
Step 1/4 : FROM nginx:1.23-alpine
--> 8e75cbc5b25c
Step 2/4 : COPY build /usr/share/nginx/html
--> 754815867973
Step 3/4 : EXPOSE 80
--> Running in 5261a5d817d3
--> Removed intermediate container 5261a5d817d3
--> 80065fec5be8
Step 4/4 : CMD ["nginx", "-g", "daemon off;"]
--> Running in 32ee7979b9ad
--> Removed intermediate container 32ee7979b9ad
--> 96f4f3656d2d
Successfully built 96f4f3656d2d
Successfully tagged vigneshmarimuthu06/dev:latest
Push to Docker Hub (dev)
The push refers to repository [docker.io/vigneshmarimuthu06/dev]
7a273e7c5148: Preparing
31531248c7cb: Preparing
f9cb3f1fd3d: Preparing
f0fb842dea41: Preparing
c1cd5c8c68ef: Preparing
1d54586a1706: Preparing
1003ff723696: Preparing
f1417ff83b31: Preparing
1d54586a1706: Waiting
1003ff723696: Waiting
f1417ff83b31: Waiting
c1cd5c8c68ef: Mounted from library/nginx
31531248c7cb: Mounted from library/nginx
f0fb842dea41: Mounted from library/nginx
f9cb3f1fd3d: Mounted from library/nginx
1d54586a1706: Mounted from library/nginx
1003ff723696: Mounted from library/nginx
f1417ff83b31: Mounted from library/nginx
7a273e7c5148: Pushed
latest: digest: sha256:119445e26db2e8e9dcea520fcf1c54fc966165076381767fe33d1f6b75d0dd7e size: 1991
✔ Build and push completed successfully!
Finished: SUCCESS
```

2. prod-deploy (triggered on pushes to main/merge)

- Build production image, push to vigneshmarimuthu06/prod (private), pull it on the production server and run it (replace old container).



prod-deploy Config [Jenkins]

3.110.176.138:8080/job/prod-deploy/configure

Incognito

Dashboard > prod-deploy > Configuration

Configure

General

Source Code Management

Build Triggers

Build Environment

Build Steps

Post-build Actions

Build Environment

☐ Delete workspace before build starts

☒ Use secret text(s) or file(s) ?

Bindings

Username and password (separated) ?

Username Variable ?

DOCKER_USER

Password Variable ?

DOCKER_PASS

Credentials ?

☒ Specific credentials ☐ Parameter expression

vigneshmarimuthu06/*****

prod-deploy [Jenkins]

3.110.176.138:8080/job/prod-deploy/

Incognito

Dashboard > prod-deploy >

Status

Changes

Workspace

Build Now

Configure

Delete Project

GitHub Hook Log

Rename

prod-deploy

Permalinks

- Last build (#4), 17 sec ago
- Last stable build (#4), 17 sec ago
- Last successful build (#4), 17 sec ago
- Last failed build (#3), 4 min 34 sec ago
- Last unsuccessful build (#3), 4 min 34 sec ago
- Last completed build (#4), 17 sec ago

Build History

trend

Filter...

#4

Oct 11, 2025, 7:59 PM

prod-deploy #4 Console [Jenki] x +

Not secure 3.110.176.138:8080/job/prod-deploy/4/console

Jenkins

Search (CTRL+K)

admin log out

Dashboard > prod-deploy > #4 > Console Output

Status

Changes

Console Output

Edit Build Information

Delete build '#4'

Timings

Git Build Data

Previous Build

Console Output

Started by user admin

Running as SYSTEM

Building in workspace /var/jenkins_home/workspace/prod-deploy

The recommended git tool is: NONE

No credentials specified

> git rev-parse --resolve-git-dir /var/jenkins_home/workspace/prod-deploy/.git # timeout=10

Fetching changes from the remote Git repository

> git config remote.origin.url https://github.com/vigneshmarimuthu/react-cicd-project # timeout=10

Fetching upstream changes from https://github.com/vigneshmarimuthu/react-cicd-project

> git --version # timeout=10

> git --version # 'git version 2.39.5'

> git fetch --tags --force --progress -- https://github.com/vigneshmarimuthu/react-cicd-project +refs/heads/*:refs/remotes/origin/* # timeout=10

> git rev-parse refs/remotes/origin/main{commit} # timeout=10

Checking out Revision 9310dcfe3c2b2aa05c5fff4bba72b87f9ed5ad42 (refs/remotes/origin/main)

> git config core.sparsecheckout # timeout=10

> git checkout -f 9310dcfe3c2b2aa05c5fff4bba72b87f9ed5ad42 # timeout=10

Commit message: "Initial commit: add Dockerfile and scripts"

> git rev-list --no-walk 9310dcfe3c2b2aa05c5fff4bba72b87f9ed5ad42 # timeout=10

[prod-deploy] \$ /bin/sh -xe /tmp/jenkins9182476462316312593.sh

+ set -e

+ DOCKER_USER=vigneshmarimuthu06

• prod-execute-shell

```
#!/bin/bash
set -e

DOCKER_USER="vigneshmarimuthu06"
PROD_IMAGE="$DOCKER_USER/prod:latest"
CONTAINER_NAME="react_prod_app"

echo "=== Login to Docker Hub ==="
echo $DOCKER_PASS | docker login -u $DOCKER_USER --password-stdin

echo "=== Build & Push Production Image ==="
docker build -t $PROD_IMAGE .
docker push $PROD_IMAGE

echo "=== Stop any container using port 80 ==="
old_port_container=$(docker ps -q --filter "publish=80")
if [ -n "$old_port_container" ]; then
    echo "Port 80 in use by container: $old_port_container - stopping it..."
    docker stop $old_port_container && docker rm $old_port_container
fi

echo "=== Stop and remove existing prod container ==="
docker ps -a -q -f name=$CONTAINER_NAME | grep -q . && docker stop $CONTAINER_NAME && docker rm $CONTAINER_NAME || echo "No old prod container found."

echo "=== Pull & Run New Production Container ==="
docker pull $PROD_IMAGE
docker run -d -p 80:80 --name $CONTAINER_NAME $PROD_IMAGE
```


- I have downloaded prod-deploy-console-output

```
Started by user admin
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/prod-deploy
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/jenkins_home/workspace/prod-deploy/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/vigneshmarimuthu/react-cicd-project # timeout=10
Fetching upstream changes from https://github.com/vigneshmarimuthu/react-cicd-project
> git --version # timeout=10
> git --version # 'git version 2.39.5'
> git fetch --tags --force --progress -- https://github.com/vigneshmarimuthu/react-cicd-project +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 9310dcfe3c2b2aa05c5fff4bba72b87f9ed5ad42 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 9310dcfe3c2b2aa05c5fff4bba72b87f9ed5ad42 # timeout=10
Commit message: "Initial commit: add Dockerfile and scripts"
> git rev-list --no-walk 9310dcfe3c2b2aa05c5fff4bba72b87f9ed5ad42 # timeout=10
[prod-deploy] $ /bin/sh -xe /tmp/jenkins9182476462316312593.sh
+ set -e
+ DOCKER_USER=vigneshmarimuthu06
+ PROD_IMAGE=vigneshmarimuthu06/prod:latest
+ CONTAINER_NAME=react_prod_app
+ echo === Login to Docker Hub ===
=== Login to Docker Hub ===
+ echo ****
+ docker login -u vigneshmarimuthu06 --password-stdin
Login Succeeded
+ echo === Build & Push Production Image ===
=== Build & Push Production Image ===
+ docker build -t vigneshmarimuthu06/prod:latest .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  2.623MB
Step 1/4 : FROM nginx:1.23-alpine
--> 8e75cbc5b25c
Step 2/4 : COPY build /usr/share/nginx/html
--> Using cache
```

```
--> Using cache
--> 754815867973
Step 3/4 : EXPOSE 80
--> Using cache
--> 80065fec5be8
Step 4/4 : CMD ["nginx", "-g", "daemon off;"]
--> Using cache
--> 96f4f3656d2d
Successfully built 96f4f3656d2d
Successfully tagged vigneshmarimuthu06/prod:latest
+ docker push vigneshmarimuthu06/prod:latest
The push refers to repository [docker.io/vigneshmarimuthu06/prod]
7a273e7c5148: Preparing
31531248c7cb: Preparing
f9cb3f1fd3d: Preparing
f0fb842dea41: Preparing
c1cd5c8c68ef: Preparing
1d54586a1706: Preparing
1003ff723696: Preparing
f1417ff83b31: Preparing
1d54586a1706: Waiting
1003ff723696: Waiting
f1417ff83b31: Waiting
7a273e7c5148: Layer already exists
f9cb3f1fd3d: Layer already exists
c1cd5c8c68ef: Layer already exists
31531248c7cb: Layer already exists
f0fb842dea41: Layer already exists
1d54586a1706: Layer already exists
f1417ff83b31: Layer already exists
1003ff723696: Layer already exists
latest: digest: sha256:119445e26db2e8e9dcea520fc1c54fc966165076381767fe33d1f6b75d0dd7e size: 1991
+ echo === Stop any container using port 80 ===
=== Stop any container using port 80 ===
+ docker ps -q --filter publish=80
+ old_port_container=
+ [ -n ]
+ echo === Stop and remove existing prod container ===
=== Stop and remove existing prod container ===
+ docker ps -a -q -f name=react_prod_app
+ grep -q .
+ echo No old prod container found.
No old prod container found.
+ echo === Pull & Run New Production Container ===
=== Pull & Run New Production Container ===
+ docker pull vigneshmarimuthu06/prod:latest
latest: Pulling from vigneshmarimuthu06/prod
Digest: sha256:119445e26db2e8e9dcea520fc1c54fc966165076381767fe33d1f6b75d0dd7e
Status: Image is up to date for vigneshmarimuthu06/prod:latest
docker.io/vigneshmarimuthu06/prod:latest
+ docker run -d -p 80:80 --name react_prod_app vigneshmarimuthu06/prod:latest
9a0d2712cd9150fdd1a6dbdd409d32bd55b7a2587740e20a2b1abc03ba1dccc4
Finished: SUCCESS
```

```
+ echo === Stop and remove existing prod container ===
=== Stop and remove existing prod container ===
+ docker ps -a -q -f name=react_prod_app
+ grep -q .
+ echo No old prod container found.
No old prod container found.
+ echo === Pull & Run New Production Container ===
=== Pull & Run New Production Container ===
+ docker pull vigneshmarimuthu06/prod:latest
latest: Pulling from vigneshmarimuthu06/prod
Digest: sha256:119445e26db2e8e9dcea520fc1c54fc966165076381767fe33d1f6b75d0dd7e
Status: Image is up to date for vigneshmarimuthu06/prod:latest
docker.io/vigneshmarimuthu06/prod:latest
+ docker run -d -p 80:80 --name react_prod_app vigneshmarimuthu06/prod:latest
9a0d2712cd9150fdd1a6dbdd409d32bd55b7a2587740e20a2b1abc03ba1dccc4
Finished: SUCCESS
```

✓ 10. Test the pipeline

- I tested the pipeline by changing the README file to “test 1,” and it triggered my pipeline automatically. The webhook and trigger are working fine.

```
ubuntu@ip-172-31-12-64:~/devops-build$ # ensure on dev branch
git checkout dev
# small change:
echo "// test change" >> README.md
git add README.md
git commit -m "test1: trigger dev pipeline"
git push origin dev
Already on 'dev'
Your branch is up to date with 'origin/dev'.
[dev 3912ee7] test1: trigger dev pipeline
Committer: Ubuntu <ubuntu@ip-172-31-12-64.ap-south-1.compute.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

1 file changed, 1 insertion(+)
Username for 'https://github.com': vigneshmarimuthu
Password for 'https://vigneshmarimuthu@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 303 bytes | 303.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/vigneshmarimuthu/react-cicd-project.git
   30ce8dd..3912ee7  dev -> dev
ubuntu@ip-172-31-12-64:~/devops-build$
```

The screenshot shows the Jenkins web interface in a browser. The address bar indicates the URL is `3.110.176.138:8080/job/dev-build/4/changes`. The Jenkins logo and navigation bar are at the top. The left sidebar contains a list of links: Status, Changes (selected), Console Output, Edit Build Information, Delete build '#4', Polling Log, Timings, Git Build Data, and Previous Build. The main content area is titled 'Changes' with a green checkmark icon. Under the 'Summary' section, it lists a change: '1. test1: trigger dev pipeline (details)'. Below this, a commit box shows 'Commit 3912ee7fb198688d35dc891c56d318f16500b01 by ubuntu' with the message 'test1: trigger dev pipeline'. At the bottom of the commit box, there is a link to view the 'README.md (diff)'.

github.com/vigneshmarimuthu/react-cicd-project

vigneshmarimuthu / react-cicd-project

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

react-cicd-project Public

dev had recent pushes 2 seconds ago Compare & pull request

main 2 Branches 0 Tags

Go to file Add file Code

Ubuntu Initial commit: add Dockerfile and scripts 9310dcf · 48 minutes ago 2 Commits

build	Add files via upload	2 years ago
.dockerignore	Initial commit: add Dockerfile and scripts	48 minutes ago
.gitignore	Initial commit: add Dockerfile and scripts	48 minutes ago
Dockerfile	Initial commit: add Dockerfile and scripts	48 minutes ago
build.sh	Initial commit: add Dockerfile and scripts	48 minutes ago
deploy.sh	Initial commit: add Dockerfile and scripts	48 minutes ago
docker-compose.yml	Initial commit: add Dockerfile and scripts	48 minutes ago

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

github.com/vigneshmarimuthu/react-cicd-project/pull/1

vigneshmarimuthu / react-cicd-project

Code Issues Pull requests 1 Actions Projects Wiki Security Insights Settings

Dev #1

Open vigneshmarimuthu wants to merge 2 commits into main from dev

Conversation 0 Commits 2 Checks 0 Files changed 1 +2 -0

vigneshmarimuthu commented now

No description provided.

Ubuntu added 2 commits 23 minutes ago

test: trigger dev pipeline 38ce8dd

test: trigger dev pipeline 3912ee7

No conflicts with base branch

Merging can be performed automatically.

Reviewers

No reviews

Still in progress? Convert to draft

Assignees

No one—assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

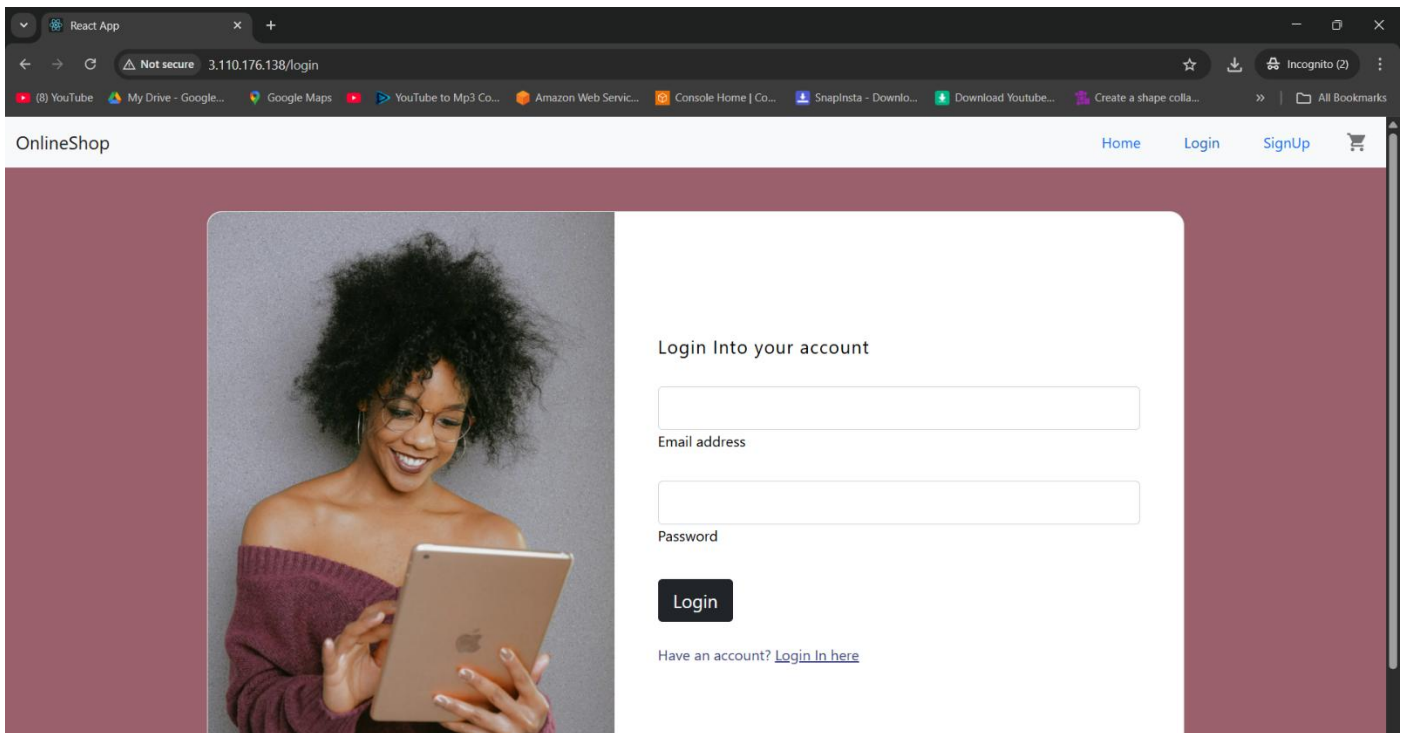
```
ubuntu@ip-172-31-12-64:~/devops-build$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	NAMES	CREATED	STATUS	PORTS
9a0d2712cd91	vigneshmarimuthu06/prod:latest	"/docker-entrypoint..."	react_prod_app	6 minutes ago	Up 6 minutes	0.0.0.0:80->80/tcp, [::]:80->80/tcp
9d42e4c5a847	jenkins/jenkins:lts-jdk11	"/usr/bin/tini -- /u..."	jenkins	23 minutes ago	Up 23 minutes	0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp

```
ubuntu@ip-172-31-12-64:~/devops-build$
```

28°C Partly cloudy

1:36 AM 10/12/2025

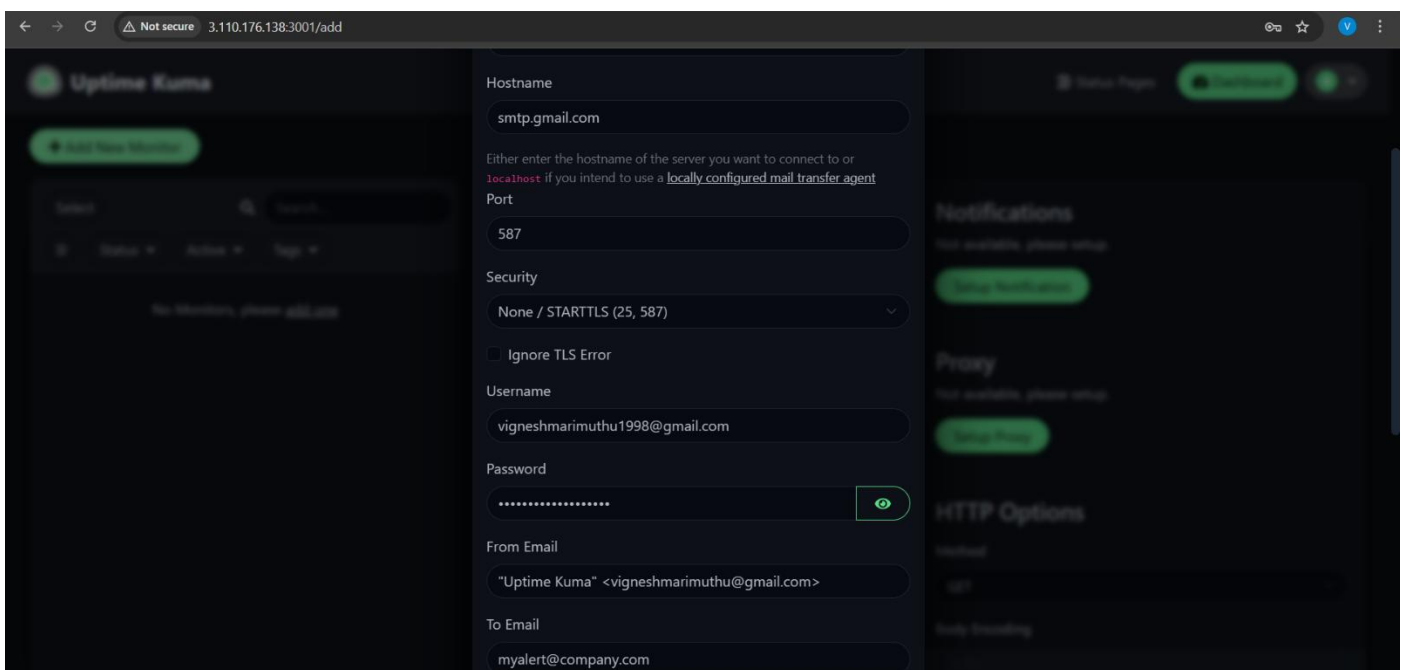
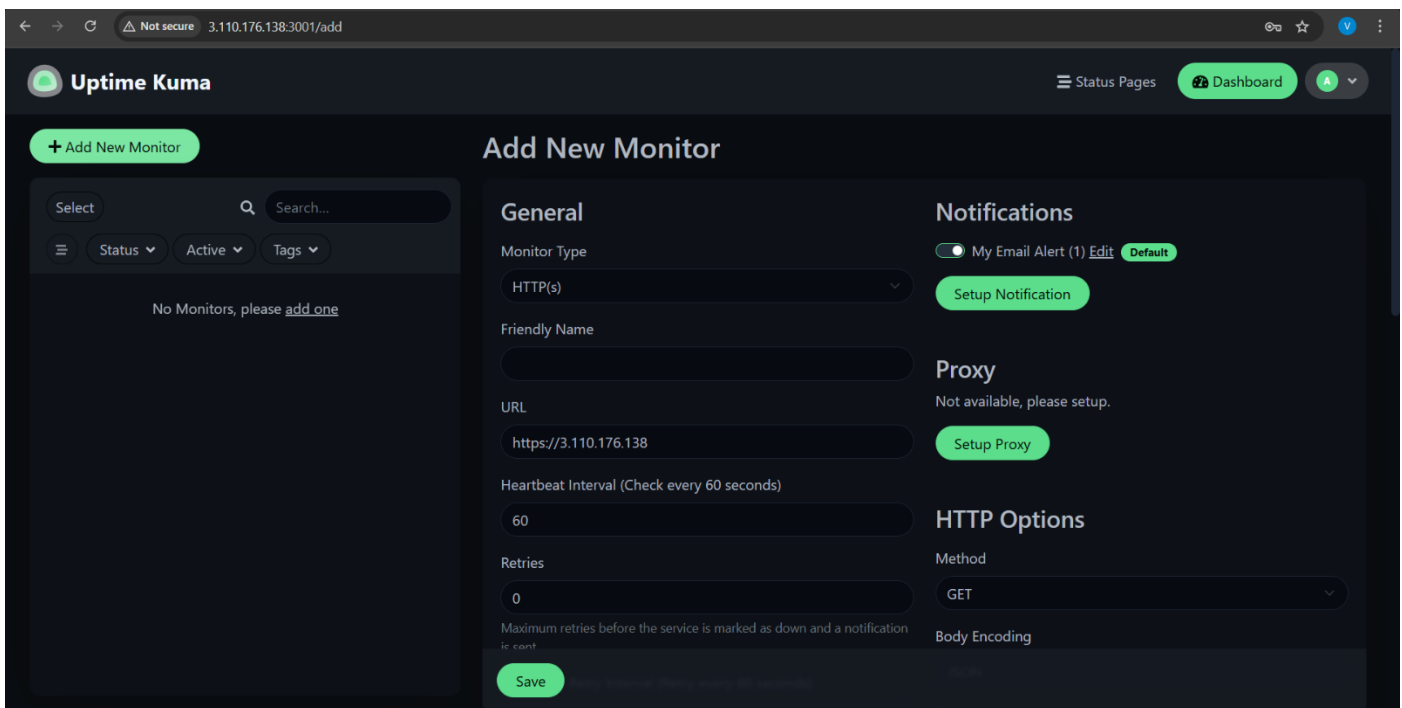
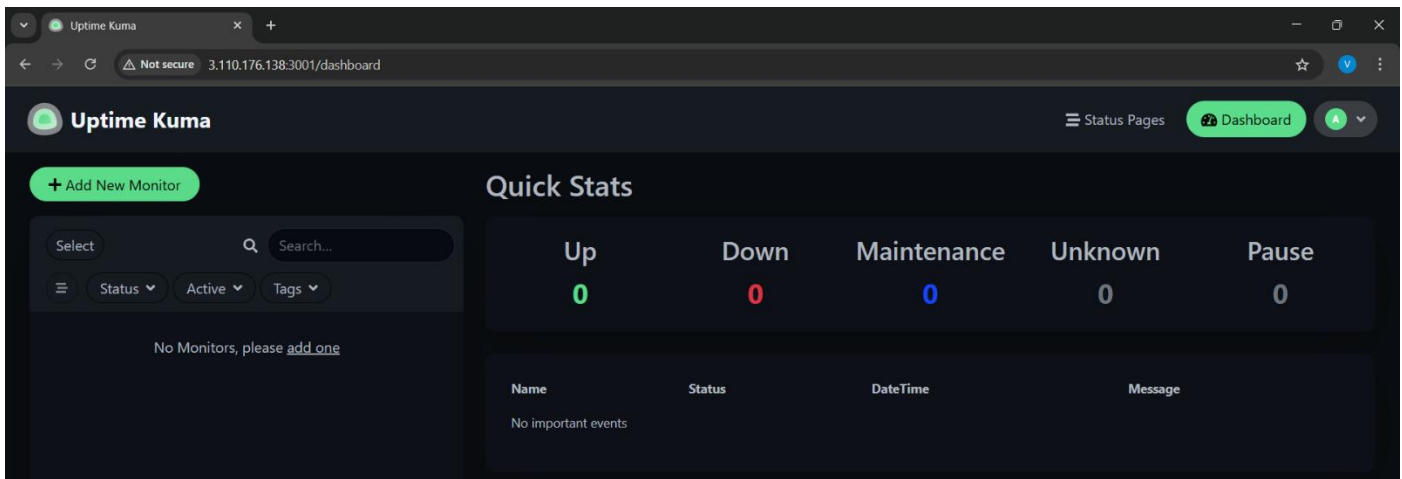


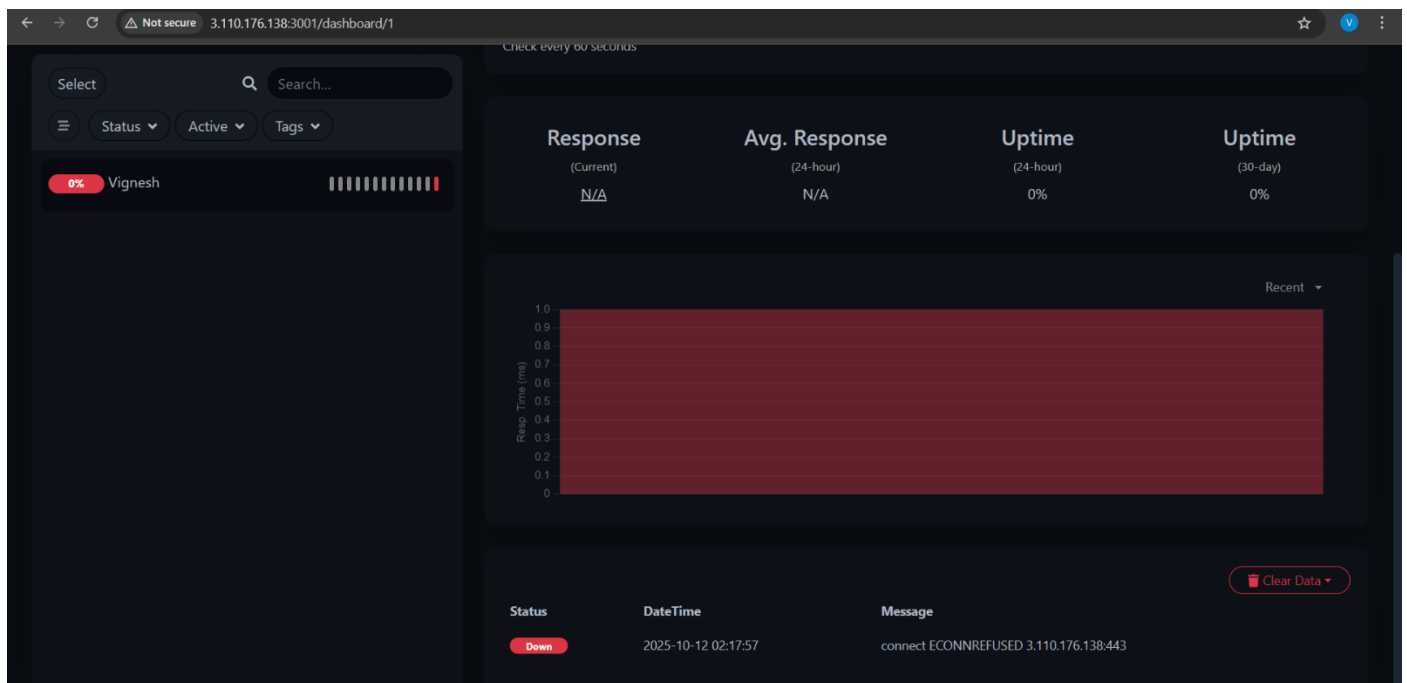
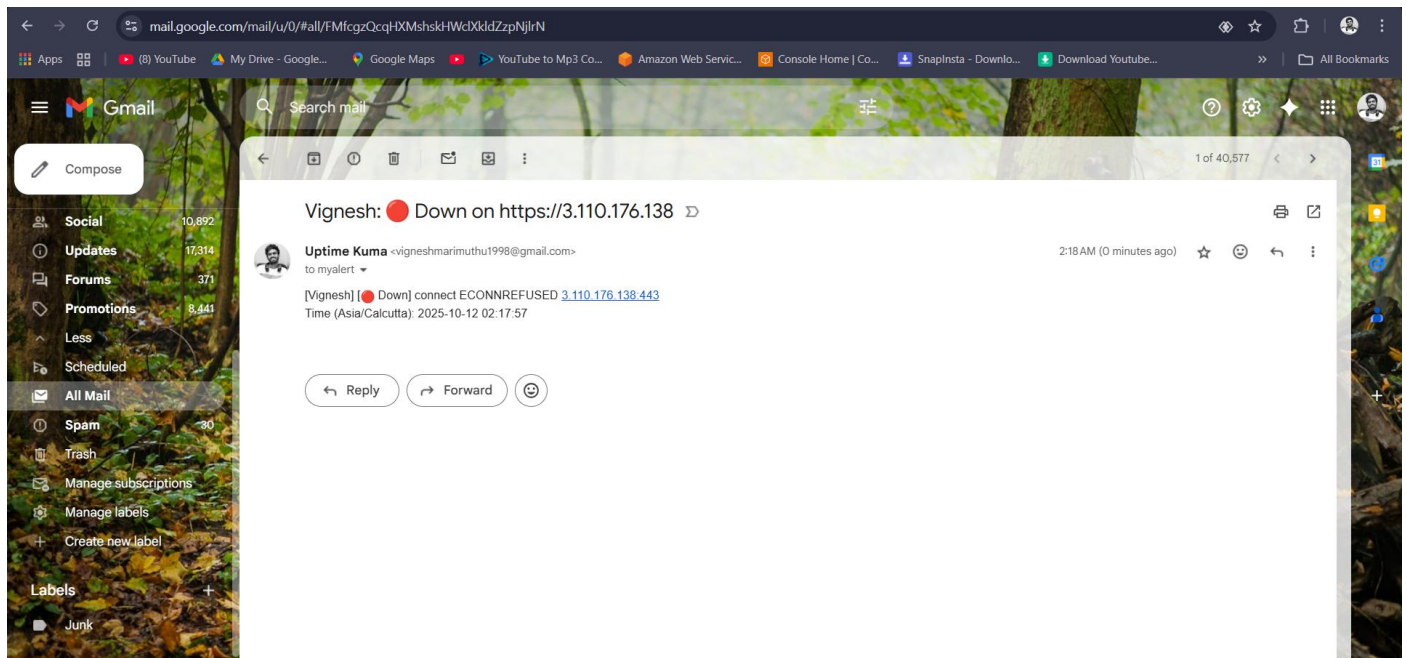
✅ 11. Monitoring: Install Uptime Kuma (self-hosted)

- Uptime Kuma is a simple, open-source uptime monitor with alerts. It will only notify you when the app is down.

```
ubuntu@ip-172-31-12-64:~/devops-bulls$ mkdir -p ~/uptime-kuma && cd ~/uptime-kuma
cat > docker-compose.yml <<'EOF'
version: '3.8'
services:
  uptime-kuma:
    image: louislam/uptime-kuma:1
    container_name: uptime-kuma
    ports:
      - "3001:3001"
    volumes:
      - ./data:/app/data
    restart: always
EOF

docker compose up -d
WARN[0000] /home/ubuntu/uptime-kuma/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid poten
tial confusion
[+] Running 13/13
✔ uptime-kuma Pulled 19.4s
✔ b338562f40a7 Pull complete 2.8s
✔ 874bf4d93720 Pull complete 2.8s
✔ b16337721583 Pull complete 4.8s
✔ 7d955db85b85 Pull complete 4.9s
✔ 2c706596bd17 Pull complete 5.0s
✔ 88a5c59ed14f Pull complete 5.0s
✔ 5a1d0a896c33 Pull complete 6.6s
✔ e68c2f25b946 Pull complete 7.0s
✔ 2e6c90f010d6 Pull complete 7.0s
✔ ff15b10fabbb Pull complete 7.1s
✔ 4f4fb700ef54 Pull complete 7.1s
✔ d2a400cc8adb Pull complete 16.1s
[+] Running 2/2
✔ Network uptime-kuma_default Created 0.1s
✔ Container uptime-kuma Started 1.6s
ubuntu@ip-172-31-12-64:~/uptime-kuma$
```





- I checked Uptime Kuma and monitored it.
- When I shut down Docker, it went down and I received a notification from the server. Everything was checked and it's all working.